



AERO4600 Automatic Flight Control Systems

Assignment 3 — 2024

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Abstract

This Report examines the lateral dynamics and gust response of an aircraft and builds on prior work ?, focusing on utilizing Matlab System Identification Toolbox as well as simulating aircraft response to gust inputs. The primary lateral modes analysed in this report are the side-slip (β) and bank angle (ϕ) of the aircraft, and are evaluated through the influence of aileron/rudder control inputs and gust disturbances. This report also evaluates the impact of sensor noise on the the system identification model and methods of improving the robustness of the system identification process. Additionally, the gust/turbulence was modelled using the Dryden Spectra method to aid in simulating the gust responses, in order to compute the maximum side-slip and bank angle perturbations. These finding contribute to modelling and understanding the aircraft's dynamics and control in the lateral motion.

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1. Introduction

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2. Q1 – Vertical Speed Transfer Function

3. Q2 – Pitch Rate Autopilot

4. Q3 – Vertical Speed Guidance Loop

The vertical speed guidance loop is designed to control the aircraft's ascent and descent rates by incorporating a pitch rate controller as an inner loop. This loop includes a compensator in the forward leg, which processes the vertical speed error to generate a pitch rate command. The pitch rate command then feeds into the inner loop to modulate the elevator input, allowing the system to adjust vertical speed in response to deviations from the desired reference.

4.1. Loop Topology

The complete loop topology of the vertical speed guidance loop is shown in figure 1. The compensator transforms the vertical speed error ($vs_{ref} - vs$) into a pitch rate command. This command is then processed within the inner loop section ($de2q$), where the transfer functions G_1 and G_2 are used in the pitch rate controller to compute the required elevator deflection (δ_e) to achieve the target pitch rate.

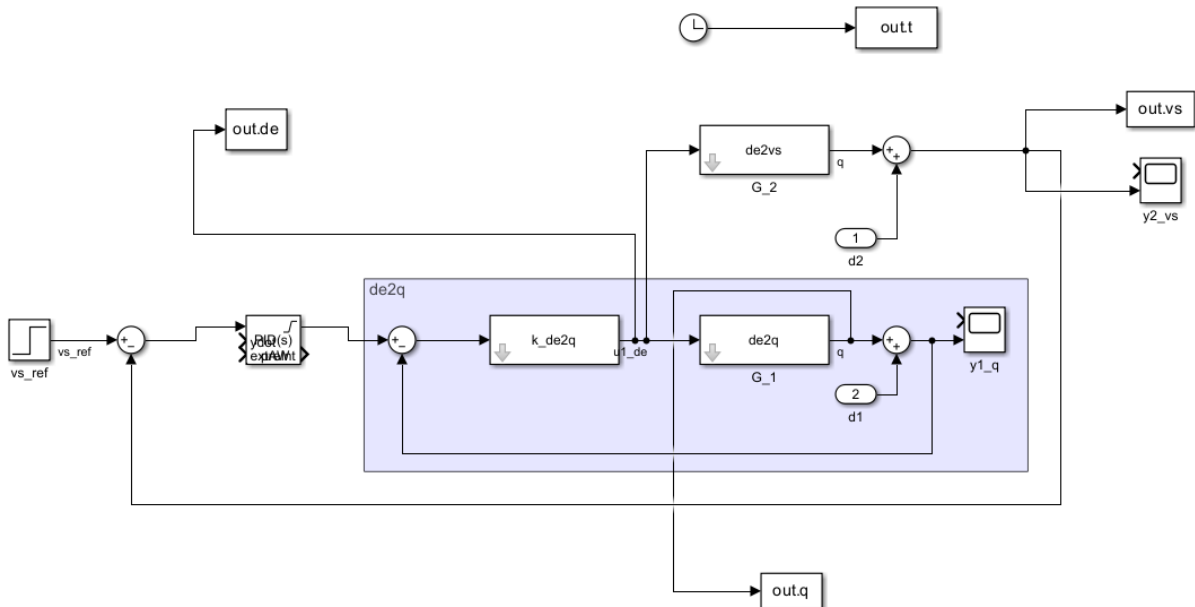


Figure 1: Vertical Speed Guidance Loop Topology

4.2. Aircraft Response to Vertical Speed Command

4.3. Pitch Rate Response

4.4. Elevator Activity

5. Q4 – Design Auto-Throttle Control System

5.1. MIMO Loop Topology

5.2. Guidance Loop Topology Specifications

5.3. Primary Control Effects

5.4. Cross-coupling Control Effects

5.5. Elevator and Throttle Response

5.6. Auto-Throttle Effectiveness

6. Final Design

6.1. Q5 – Implementation of the control and guidance laws

6.2. Q6 – Gust Effectiveness

6.3. Q7 – Commanded Large Vertical Speed

7. Conclusion

References

A. Example of a Table

Table 1: Bachelor of Engineering Mechatronics First Year Program

1st Year		
Semester	Course Code	Course Name
1	ENGG1003	Introduction to Procedural Programming
1	ENGG1500	Introduction to Professional Engineering
1	MATH1110	Maths for Engineering, Science & Technology 1
1	MECH1110	Mechanical Drawing/CAD & Workshop Practice
2	CIVL1100	Fundamentals of Engineering Mechanics
2	ELEC1310	Introduction to Electrical Engineering
2	MATH1120	Maths for Engineering, Science & Technology 2
2	MECH1750	Engineering Materials I